

Schur Rings Over Numerical Semigroups

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1. Numerical Semigroups
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Numerical Semigroups

All information on numerical semigroups is sourced from "Numerical Semigroups and Applications" [Abdallah Assi,]

A numerical semigroup is a semigroup with addition which is a subset of the natural numbers.

e.g

- $\mathbb{N}$
- $\mathbb{Z}^+$
- $n\mathbb{Z}^+, \forall n \in \mathbb{Z}^+$
- $\{3, 5, 6, 8, 9, 10, 11, \rightarrow\}$

Properties of Numerical Semigroups

Any numerical semigroup can be reduced to a finite generating set.

e.g.

- $\mathbb{Z}^+ = \langle 1 \rangle$
- $n\mathbb{Z}^+ = \langle n \rangle$
- $\{3, 5, 6, 8, 9, 10, 11, \rightarrow\} = \langle 3, 5 \rangle$

It can be assumed the gcd of a generating set is 1.

Properties of Numerical Semigroups, Cont.

A numerical semigroup with at least two generators greater than 0 has a Frobenius number and a conductor.

The Frobenius number is the greatest gap, e.g. $F(\langle 3, 5 \rangle) = 7$

The conductor is the next natural number, e.g. $C(\langle 3, 5 \rangle) = F(\langle 3, 5 \rangle) + 1 = 8$

Partitions on Schur Rings

Let $S = \langle x^3, x^5 \rangle$. Consider the polynomial ring $\mathbb{Z}[S]$.

The coarsest Schur ring is the ring such that if $a, b \in S$ always have the same coefficient in $\mathbb{Z}[S]$, then they are "combined".

Partitions on Schur Rings Cont.

Start by combining the generators, $x^3 + x^5$.
Then check higher orders.

$$(x^3 + x^5)^2 = x^6 + 2x^8 + x^{10}$$

There is no other way to generate x^6 or x^{10} , so they can be combined.
Similarly, there is no other way to generate x^8 , so it is a singleton.

Partitions on Schur Rings Cont.

$$(x^3 + x^5)^3 = x^9 + 3x^{11} + 3x^{13} + x^{15}$$

$x^{15} \in S^5, x^9 \notin S^5$, thus they are both singletons.

x^{11}, x^{13} ? They both can only be generated by $x^8(x^3 + x^5)$, so they can be combined.

No further combining is possible.

Final Coarsest Schur Ring: $x^3 + x^5, x^6 + x^{10}, x^8, x^9, x^{11} + x^{13}, x^{12}, \rightarrow$

Finer Schur rings are possible.

Primary Research Question

Conjecture

All Schur Rings over numerical semigroups are eventually discrete.

Theorems

Theorem 1

For $S = \langle p, q \rangle$ where $p < q$ and $\gcd(p, q) = 1$, no element of S greater than $(p + q)(p - 1)$ can be combined in a Schur ring.

Theorem 2

For any numerical semigroup of the form $S = \langle m, m + 1, m + 2, \dots, 2m - 1 \rangle$, any Schur ring over S is eventually discrete.

Theorem 3

For the numerical semigroup, $S = \langle a_0, a_1, \dots, a_k \rangle$, if $\gcd(a_i, a_j) = 1$ for all $0 \leq i, j \leq k$ with $i \neq j$, then any Schur ring over S is eventually discrete.

Future Objectives and Events

Eventual Goal: Find a proof for the primary conjecture.

Upcoming events:

- Presenting at UCUR
- Presenting at The Festival of Excellence.



Abdallah Assi, P. A. G.-S.

Numerical semigroups and applications.

The End